

Executive Summary

Occupational health and safety rely on integrated data and analytics to predict and prevent injuries. An AI-driven portal should ingest rich medical and workplace data (health records, job history, exposures, etc.), apply validated risk models, and present actionable reports. Key features include user-specific dashboards, explainable risk scores, alerts, and customizable reports. The system must use healthcare standards (FHIR, SNOMED CT, LOINC, RxNorm) and occupational codes (O*NET, SOC) for data exchange ¹ ². It should comply with regulations (HIPAA, GDPR) through encryption and de-identification (Safe Harbor or expert methods ³) and ensure fairness (bias testing) and auditability. Authoritative sources (OSHA/NIOSH/CDC guidelines, BLS injury statistics ⁴, clinical guidelines, workers' comp databases ⁵) will inform model training and reporting. We recommend ensemble risk models (e.g. gradient boosting) with rigorous validation (cross-validated AUC, calibration curves, subgroup analysis), continuous monitoring, and user training. The implementation roadmap spans data integration, model development, pilot testing, and deployment, with an API-driven modular architecture (see architecture diagram). Tables below compare data sources, models, and integrations.

Portal Features and Modules

- **Data Inputs:** Integrate clinical (EHR diagnoses, labs, imaging), demographic, job and exposure data (via HL7 FHIR *Occupational Data for Health* profiles ²), safety reports, wearable sensors, and external knowledge bases (PubMed/CDC guidelines).
- **Risk Models:** Use predictive models (e.g. logistic regression, Random Forest, XGBoost/LightGBM, neural nets) to estimate injury or condition-worsening likelihood ⁶. Models should output calibrated risk scores and key contributing factors.
- **User Roles:** Support roles such as occupational physicians, safety officers, case managers, and admins. Include role-based access controls and authentication ⁷. User management should allow auditing of actions (viewing/modifying data, model versions) for transparency ⁷.
- **User Interface/UX:** Provide dashboards summarizing overall risk and individual profiles. Interfaces should be intuitive, with drill-down to patient or case details. Visualizations (risk stratification charts, factor importance graphs) aid interpretation. Include templates for standardized reports (PDF or interactive) that list top risk factors and recommendations. UI design should follow usability best practices and accessibility.
- **Report Templates:** Predefined formats (e.g. summary page, detailed analysis) for clinical review. Each report should include patient/job context, risk scores, contributing variables, charts, and guidance references. Allow exporting and printing.
- **Alerts & Notifications:** Configurable thresholds trigger alerts (email/EHR alerts) when risk exceeds set limits. For example, high predicted injury risk may prompt a review. Support both real-time monitoring (streaming sensor data) and batch mode (daily or weekly analyses).
- **Explainability:** Integrate model interpretability (e.g. SHAP values, feature importance) so experts understand predictions. Provide textual explanations or counterfactuals. This enhances trust and clinical decision support.

- **Fairness/Bias Mitigation:** Implement fairness testing across subgroups (age, sex, job class, race). Use metrics like equalized odds or calibration within groups. Apply data balancing or algorithmic fairness techniques as needed.
- **Privacy/Security:** Encrypt data at rest/in transit; comply with HIPAA and GDPR. Use de-identification (Safe Harbor or expert determination ³) for research data. Audit logs must record access to protected data and model changes.
- **Auditability & Versioning:** Maintain version control for models, data schemas, and code. Log all system changes. Include an audit trail for each prediction (timestamp, model version, inputs).
- **Clinical Decision Support (CDS) Integration:** Expose predictions via standards-based APIs (FHIR, CDS Hooks). Allow the portal to call out or embed in EHR systems. The CDS component should retrieve relevant clinical guidelines and flag patients who need intervention ⁸.
- **Integration/APIs:** Use FHIR APIs and HL7 messaging to pull data from EHRs, HR systems, and safety databases ⁸ ¹. Support RESTful and HL7v2 interfaces. Offer connectors to O*NET or industry code systems for job descriptors.
- **Offline/Real-time Modes:** The portal should operate in near-real-time for urgent alerts (e.g. sensor-triggered fall risk) and in batch mode for periodic risk assessments.
- **Customization & Extensibility:** Allow threshold and model parameter configuration per organization. Enable adding new risk factors or modules. Offer role-tailored views and branding.
- **Training/Education:** Provide user guides, training modules, and documentation explaining model limitations. Educate users on interpreting risk outputs and on legal boundaries (e.g. avoiding workplace discrimination).
- **Compliance:** Ensure the system can produce audit reports to demonstrate adherence to OSHA/NIOSH recommendations. If considered Software as a Medical Device (SaMD) under FDA, follow SaMD guidance for risk scoring software ⁹. Comply with healthcare IT standards (21st Century Cures Act, etc.) for interoperability.

Data Types and Schemas

The portal requires extensive structured data. Key types include:

- **Demographics:** Age, sex, ethnicity, employment status, years on job. (Use FHIR *Patient* and *Usual Work* resources.)
- **Medical History & Comorbidities:** Diagnoses (ICD-10, SNOMED CT codes), surgery history, chronic conditions, medications (RxNorm codes), allergies. (FHIR *Condition*, *MedicationRequest*.)
- **Occupational History:** Current and past jobs with job title, industry (NAICS), essential job functions. Use HL7 FHIR ODH profiles: occupation, employer, job duties, work classification (full/part-time), work schedule ² ¹⁰. Map roles to standardized job descriptors (O*NET or SOC codes) for risk factors.
- **Exposure Metrics:** Quantitative/qualitative exposure data (chemical exposures, noise levels, vibration, ergonomic strain). Include dates and duration. (Could use FHIR *Observation* with LOINC codes for e.g. toxin levels). Use industry exposure thresholds (OSHA PELs, ACGIH TLVs).
- **Workplace Controls:** Presence of safety measures (PPE usage, training logs, ergonomic adjustments, engineering controls). (Custom FHIR *Observation* or *Procedure* profiles).
- **Incident and Injury History:** Past work-related injuries or illnesses (dates, nature, treatment, lost workdays). Include workers' comp claims data: injury nature, cause, medical cost, wage loss ⁵. Also near-misses or safety violations from incident tracking systems.

- **Functional Assessments:** Results of physical/functional tests (e.g. lifting tests, mobility exams, FCE scores), occupational health exam results. (FHIR *Observation*, *Procedure*.)
- **Laboratory and Imaging:** Lab results (blood tests, etc.) coded in LOINC. Imaging findings (X-rays, MRIs) with reports (FHIR *ImagingStudy*, DICOM data). These may reveal chronic conditions affecting injury risk.
- **Time Series Data:** Temporal data such as wearables (heart rate, motion data), repeated symptom surveys, or longitudinal test results. (FHIR *Observation* series.) Model can use trajectories/trends.
- **Environmental/Contextual Data:** Job environment factors (shift length, team size, weather if outdoor, etc.), socioeconomic factors if relevant. (Custom or FHIR *Observation*).
- **Schema and Standards:** Data should align with US Core Data for Interoperability (USCDI) and relevant FHIR profiles. Terminologies: SNOMED CT/ICD-10 for diagnoses, CPT/HCPCS for procedures, LOINC for labs, RxNorm for meds, O*NET/SOC for occupations, WCIO codes for injury details ¹.

Authoritative Sources and Data Sets

We recommend integrating or referencing these primary sources (with URLs):

Source / Database	Description / Use	Type
OSHA (osha.gov)	Federal workplace safety regs and guidance, Industry-specific standards. Injury/illness reporting rules.	Government, Regulatory
NIOSH/CDC (cdc.gov/niosh)	Occupational safety research, surveillance data (NIOSH publications, FACE reports), NIOSH pocket guides. State surveillance systems (NIOSH-funded).	Government (Research)
CDC (cdc.gov)	Public health guidelines, injury surveillance (WISQARS), CDC/NIOSH bulletins on worker health. National guidelines (e.g. pandemic return-to-work).	Government
BLS (bls.gov)	National injury stats: SOII (Survey of Occupational Injuries/Illnesses), CFOI (Fatal Injuries), OIICS coding. [Economics Daily] ⁴ . Data by industry, occupation.	Government (Statistics)
WHO Occupational Health (who.int)	Global occupational health guidelines, Hazard classifications, ILO/WHO international standards.	International Agency
PubMed/NIH PMC (ncbi.nlm.nih.gov)	Peer-reviewed literature on occupational risk factors, predictive models, systematic reviews.	Research Literature
ClinicalTrials.gov	Ongoing clinical or intervention trials on occupational health.	Research Registry

Source / Database	Description / Use	Type
Workers' Comp Databases (e.g. NCCI, WCRI)	Claims data (injury type, costs, lost time). State WCIO coding info ¹¹ . Useful for cost/outcome modeling.	Industry/ Research
National Injury Surveillance (e.g. NEISS)**	Emergency dept. injury data (CPSC), state surveillance (e.g. Michigan OSHA), poison control (for exposures).	Government (Data)
ACOEM Guidelines (acoem.org)	Evidence-based practice guidelines in occupational medicine (e.g. disability management).	Professional Organization
CDC/NIOSH Databases (injuryfacts.nsc.org)**	Injury Facts (NSC) for national trends; CDC WONDER/WISQARS for ER/hospital data.	Aggregator
O*NET (onetonline.org)	Standardized job characteristics (physical demands, hazards per occupation). Used for exposure assessment.	Government (Labor)
SNOMED CT (snomed.org)	Clinical terminology for conditions and findings.	Terminology Authority
ICD-10 (who.int/classifications/icd)	Diagnosis coding (often used in EHR).	Terminology Authority
LOINC (loinc.org)	Standard codes for lab tests and measurements.	Terminology Authority
RxNorm (nih.gov)	Standard codes for medications.	Terminology Authority
HL7 FHIR (hl7.org/FHIR)	Healthcare data exchange standard; specifically the US Occupational Data for Health (ODH) Implementation Guide ² .	Standards Body
EHR Vendors / APIs (Epic, Cerner)	Implementation notes for integration (FHIR/HL7 support, USCDI compliance).	Vendor Documentation
ISO/IEEE/IEC Standards	AI/ML frameworks (ISO 23894 AI in healthcare, ISO 42001 AI management, IEC 62304 software, etc.) ¹² .	Standards Body
FDA Guidance	If considered SaMD: FDA's Clinical Decision Support and AI/ML device guidelines.	Government (Regulator)

All URLs above are authoritative. The portal should prioritize governmental and peer-reviewed sources. For example, OSHA/NIOSH sites are primary references. EHR and standards documentation (HL7, LOINC) ensure interoperability. PubMed and guidelines (ACOEM, ISO) inform model and system validation.

Modeling and Validation

- **Model Types:** Candidate approaches include classical (logistic regression, Cox survival models), tree-based ensembles (Random Forest, XGBoost/LightGBM), and neural networks. Time-to-event or recurrent neural nets can handle longitudinal data. Ensemble models like LightGBM have shown high accuracy on complex healthcare data ⁶. Models should output probabilities and relative risk (adjusted for covariates).
- **Training/Testing:** Use cross-validation (k-fold) or bootstrapping. Preserve temporal order when appropriate (train on past years, test on later). For fairness, ensure balanced sampling across subgroups.
- **Performance Metrics:** Report discrimination (AUROC, AUPRC), calibration (Brier score, calibration plots), and decision-curve metrics. Also report sensitivity, specificity, PPV/NPV at chosen thresholds. No single metric suffices ¹³.
- **Fairness Testing:** Evaluate model parity across groups (e.g. race, sex, age). Compute metrics like equalized odds difference or demographic parity. Mitigate bias by reweighting or adversarial methods if needed.
- **Calibration:** Perform recalibration (e.g. isotonic regression) if model over/under predicts risk. Validate calibration on hold-out data and over time.
- **External Validation:** Test models on independent datasets (e.g. different industry, region). Possibilities include separate company's claims, another state's injury logs, or published datasets. Track performance drift.
- **Monitoring:** Continuously monitor prediction accuracy (e.g. with periodically calculated AUC) and data drift (feature distribution changes). Use a model registry to track versions and retraining triggers.
- **Explainability:** Employ tools like SHAP, LIME, or integrated gradients to explain individual predictions. Provide feature attributions in reports. The system should surface top risk factors (e.g. "heavy lifting frequency" or "uncontrolled hypertension") that contributed most to the risk score.
- **Regulatory Validation:** If AI output is considered medical advice, conform to standards (ISO 62304, Good Machine Learning Practice). Conduct performance testing akin to clinical trials (sensitivity to real-world variation). Maintain documentation of validation steps and results.

Data Governance and Legal Considerations

- **Privacy & Consent:** All personal health information (PHI) must be protected. Obtain informed consent or use data under a business associate agreement. De-identify data when used for modeling (HIPAA Safe Harbor/Expert methods ³). Comply with HIPAA (US) and GDPR (EU) for cross-border data: e.g. data minimization, data subject rights.
- **Security:** Use strong access controls (role-based, MFA), encryption, and secure audit trails. Regular security audits and penetration tests should be performed.
- **Consent Management:** Keep records of patient/workers' consent for data use. Provide opt-out mechanisms if legally required.
- **Auditability:** Log all data accesses and model uses. The system should support retrospective auditing to investigate liability or bias claims.
- **Legal Risks:** Be mindful of employment law (ADA, workers' compensation statutes). Ensure risk predictions are used for health/safety, not for punitive measures. Disclaimers should state that predictions are probabilistic assessments (not diagnoses).

- **Compliance:** The portal may fall under health IT regulations. If marketed, evaluate if FDA SaMD rules apply (software giving risk probabilities is often regulated ⁹). Adhere to OSHA recordkeeping requirements and any state regulations on occupational health.

Implementation Roadmap and Architecture

A phased rollout is recommended:

- 1. Planning (3–4 months):** Gather requirements from clinicians, safety experts, and IT. Define scope of injuries/conditions to predict. Identify data sources. Conduct feasibility.
- 2. Data Integration (6–9 months):** Map data schemas (EHR, HR, incident logs) to FHIR/ODH profiles ². Build ETL pipelines. Establish API connections (FHIR interfaces, ONET feeds). *Ensure compliance (HIPAA) through encrypting pipelines.*
- 3. Model Development (6–9 months):** *Train initial models on historical data. Iteratively feature-engineer using expert input (e.g. encode heavy lifting frequency, prior claims). Use cross-validation, tune hyperparameters (e.g. in XGBoost). Implement explainability modules.*
- 4. Validation & Pilot (3–4 months):** *Test with held-out data and external sets. Refine model. Conduct a pilot within one department or site. Gather user feedback on UI and report formats. Adjust thresholds to clinical needs.*
- 5. Deployment (3–6 months):*** Roll out to broader user base. Set up continuous monitoring (performance dashboard). Provide training workshops. Schedule periodic model retraining (e.g. annually).

Below is an example architecture diagram. It shows data inflow from EHR/HR systems and external sources into a data warehouse and analytics engine, producing a risk scoring service and dashboards ⁸ ⁷.

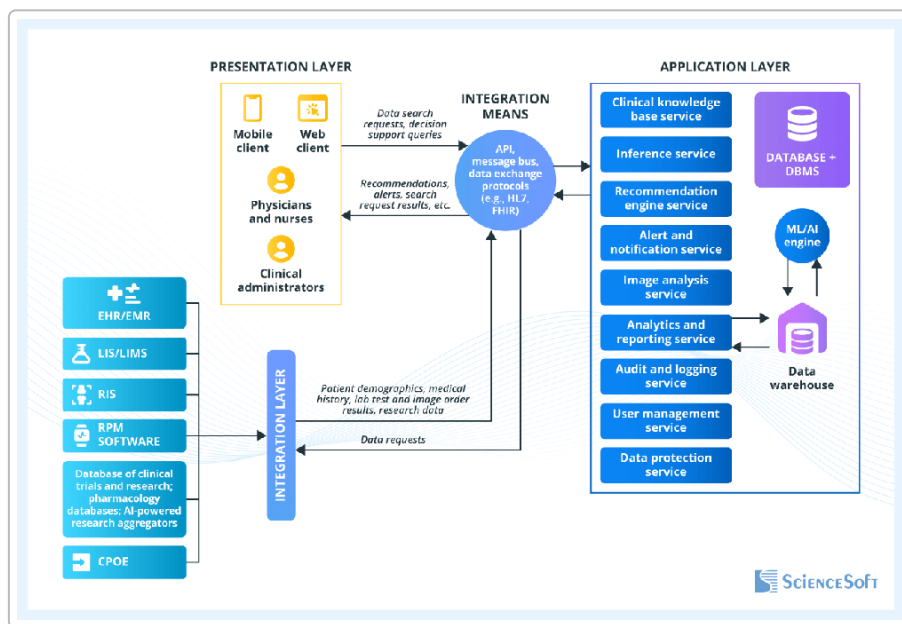


Figure: Example AI system architecture. Data from EHRs, safety logs, and guidelines feed into an ML engine. Security, audit, and user management layers ensure compliance ⁸ ⁷.

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timeline
  title Project Roadmap (2024-2025)
  2024-Q1 : Requirements gathering & design
  2024-Q2 : Data integration & pipeline setup
  2024-Q3 : Model development & iterative testing
  2024-Q4 : Pilot deployment & validation
  2025-Q1 : Expand rollout, training & monitoring
  2025-Q2 : Maintenance, regulatory review, updates

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Visualization and Reporting

- **Dashboard Views:** Show aggregate risk distributions (histograms or heatmaps by department/job), trends over time, and filter by site or job.
- **Individual Reports:** Include charts like time-series of risk score, bar chart of top contributing factors (SHAP values), and gauges for vital signs or exposures. Example charts: bar charts of risk by age group, line charts for symptom progression.
- **Color-coding:** Use risk heatmaps (green/yellow/red) for quick prioritization.
- **Report Layout:** Title and summary, followed by sections for Data Summary, Risk Score, Contributing Factors, Recommendations, and Data Sources.
- **Sample Figures:** The BLS data (Table on injury rates by age ⁴) suggests charts by age or industry. Workers' comp dashboards (e.g. CDC NIOSH WC dashboards ¹⁴) can inspire industry-sector breakdowns.
- **Tables:** Summarize risks by subgroup (e.g. Table of risk scores vs. age cohorts).
- **Comparative Tables:** Provide tabular comparisons of candidate data sources (e.g. SOII vs. workers comp), model algorithms (metrics and pros/cons), and integration options (EHR APIs vs flat file). This helps stakeholders choose approaches.

Key References

This report is informed by primary sources: OSHA/NIOSH guidelines, CDC surveillance data ¹⁵ ⁵, peer-reviewed studies on predictive models ⁶, and HL7/standards documentation ² ¹. Tables above list major data repositories and standards. All information is drawn from authoritative English-language sources; any data gaps should be filled by direct consultation with subject-matter experts.

¹ ⁷ ⁸ ⁹ ¹² Clinical Decision Support System: Features, Architecture, Costs
<https://www.scnsoft.com/healthcare/clinical-decision-support-system>

² ¹⁰ HL7.FHIR.US.ODH\Home Page - FHIR v4.0.1
<http://build.fhir.org/ig/HL7/us-odh/>

³ Guidance Regarding Methods for De-identification of Protected Health Information in Accordance with the Health Insurance Portability and Accountability Act (HIPAA) Privacy Rule | HHS.gov
<https://www.hhs.gov/hipaa/for-professionals/special-topics/de-identification/index.html>

4 There were 2.6 million nonfatal workplace injuries and illnesses in 2023 : The Economics Daily : U.S. Bureau of Labor Statistics

<https://www.bls.gov/opub/ted/2025/there-were-2-6-million-nonfatal-workplace-injuries-and-illnesses-in-2023.htm>

5 11 14 Workers' Compensation Data | Workers' Compensation | CDC

<https://www.cdc.gov/niosh/worker-comp/about/index.html>

6 An enhanced sine cosine algorithm based light gradient boosting machine for occupation injury outcome prediction | Discover Computing | Springer Nature Link

<https://link.springer.com/article/10.1007/s10791-025-09759-z>

13 On evaluation metrics for medical applications of artificial intelligence

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8993826/>

15 Nonfatal Occupational Injury Surveillance Data: Examples from Michigan | NIOSH Science Bulletin | CDC

<https://www.cdc.gov/niosh/bulletin/2024/michigan-survey.html>